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Revision History

Rev	Description	Date	Approved/Designer
X	Initial Schematic	OCT-2024	Efren D/ Antonio Q
X4	A070 Release	11/12/24	Efren D/ Antonio Q
A	BOM Optimization TPs removed (OSDA)	11/28/24	Efren D/ Antonio Q
	R17 to zero ohm resistor SW1 removed, VBUS_IN to jack (J10) connector FXLS8964AF added	12/11/24	Efren D/ Antonio Q
	Temp Sensor (U18) added	12/13/24	Efren D/ Antonio Q
	A085 Release	12/18/24	Efren D/ Antonio Q
A1	Y2 replacement	01/14/25	Efren D/ Antonio Q
AX1	Test points cleaning Added ethernet strap-in resistors Change on VBUS domain levels on CAN/LIN	05/09/25	Efren Diaz
	Rebranding name	05/20/25	
	C161, C162, R731 added	05/27/25	Efren D/ Antonio Q
	SJ6, R732, C163 added Replace SJ6 for JP1 (3x1 header)	06/06/25	
B	Release	07/08/25	Efren D/ Antonio Q
C	PTC15 connect to MII_RX_DV, PTB22 to ENET_MII_CRS	11/05/25	Efren D/ Antonio Q

S32K344MINI-EVB

" TLALOC "

Customer Evaluation Board

C A U T I O N :

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Signals (ports) have not been routed via busses as this makes it harder to determine where each signal goes.

User notes are given throughtout the schematics.

Specific PCB LAYOUT notes are detailed in ITALICS

2 Different test points used in design:

TPx - Through Hole Pad small

TPHx - Through Hole Pad Large (for standard 0.1" header). Also used on IO Matrix (IOMx)



Notes:

- All components and board processes are to be ROHS compliant
- All connectors and headers are denoted Jx/JPx and are 2.54mm pitch unless otherwise stated
- All jumpers are denoted Jx. Jumpers are 2mm pitch
- Jumper default positions are shown in the schematics. For 3 way jumpers, default is always posn 1-2.
  2 Pin jumpers generally have the "source" on pin 1.
- All switches are denoted SWx
- All test points are denoted TPx

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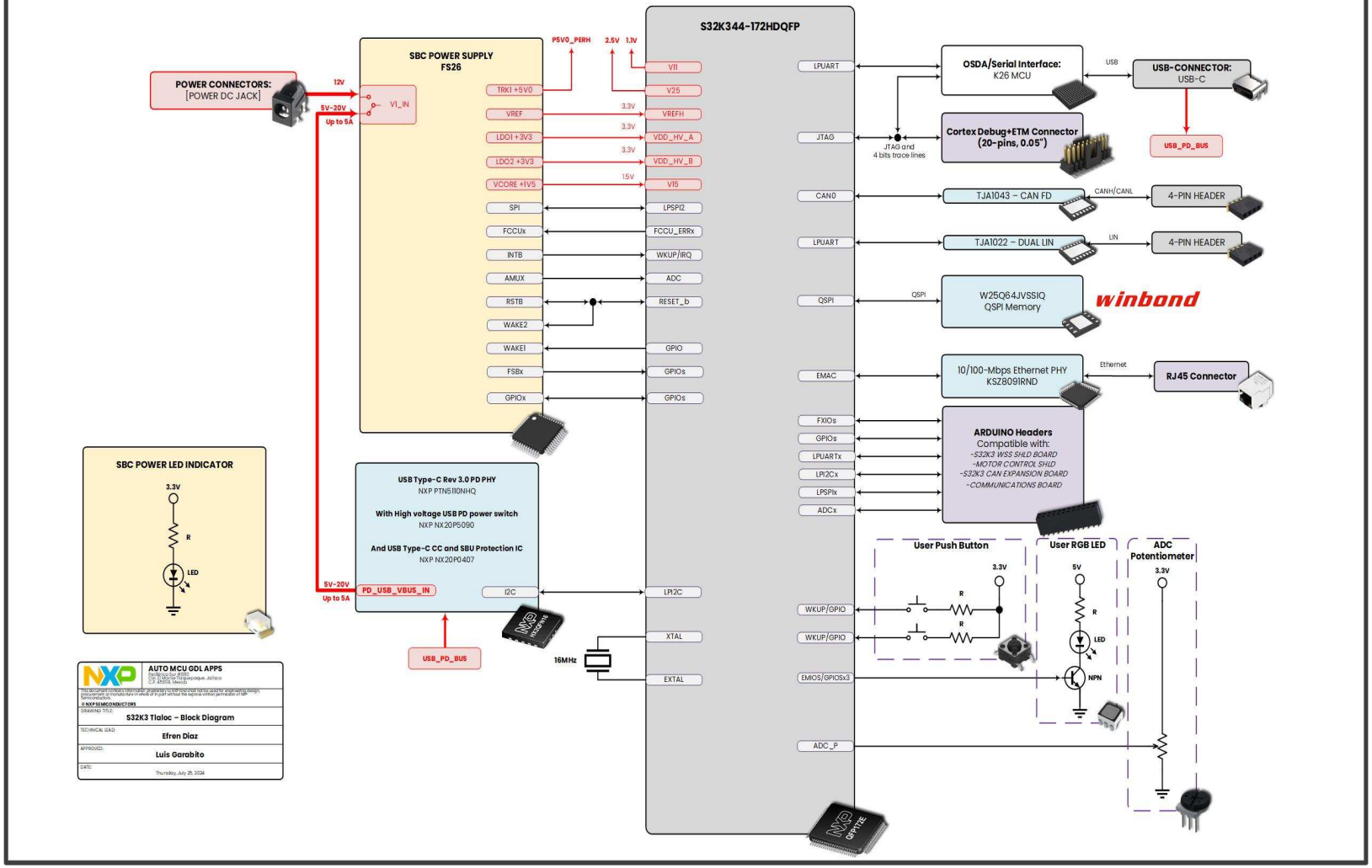
Designer: Antonio Q. / Efren D.	Drawing Title: S32K344MINI-EVB	
Drawn by: Antonio Quiroz	Page Title: TITLE PAGE	
Approved: Efren Diaz	Size C / Document Number SCH-94921 PDF: SPF-94921	Rev C
	Date: Wednesday, November 05, 2025 / Sheet 1 of 14	

S32K3 Tlaloc

S32K344MINI-EVB

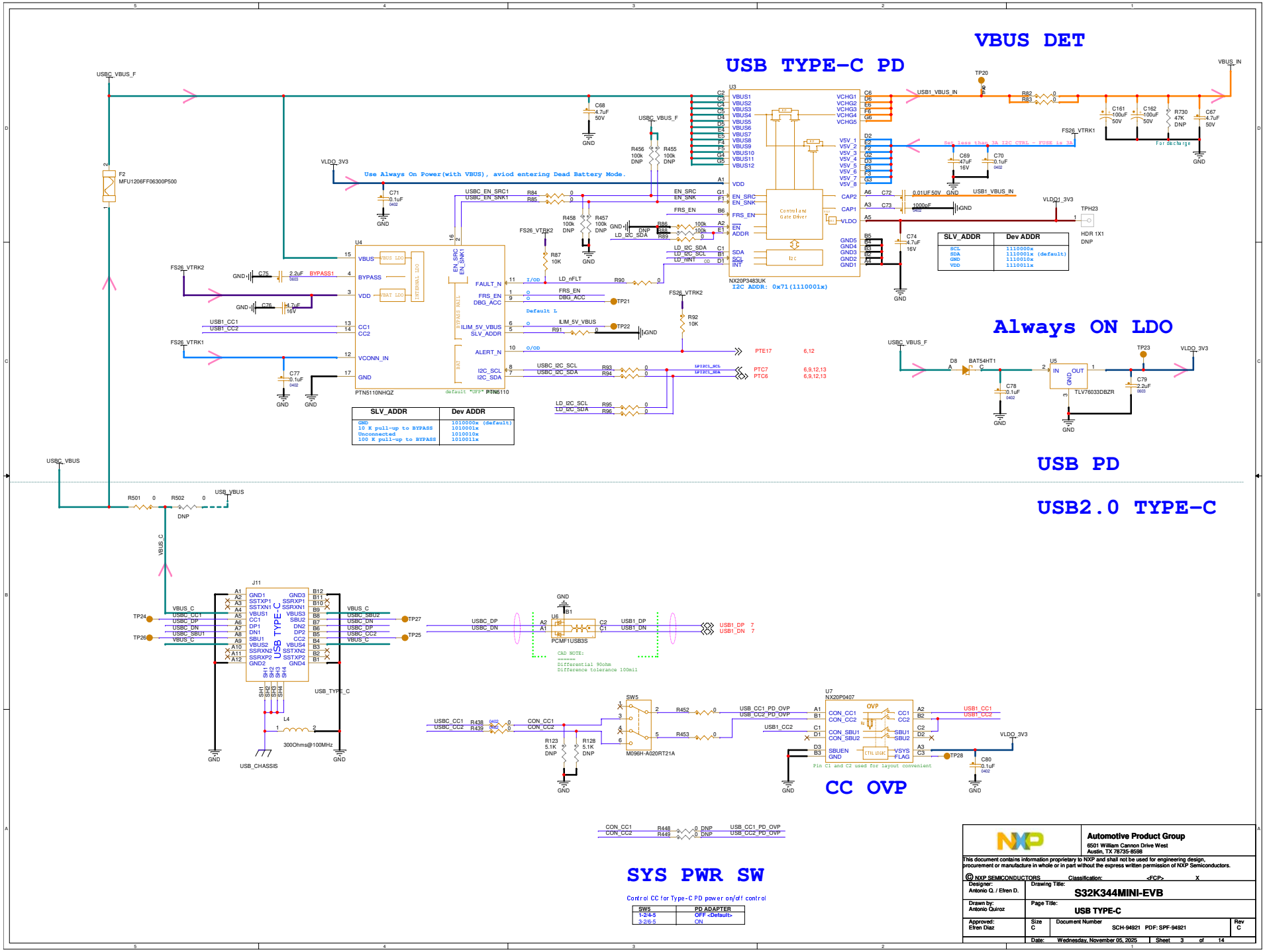
S32K344 HDQFP172 Tlaloc Evaluation Board

05/27/2024 Rev B.



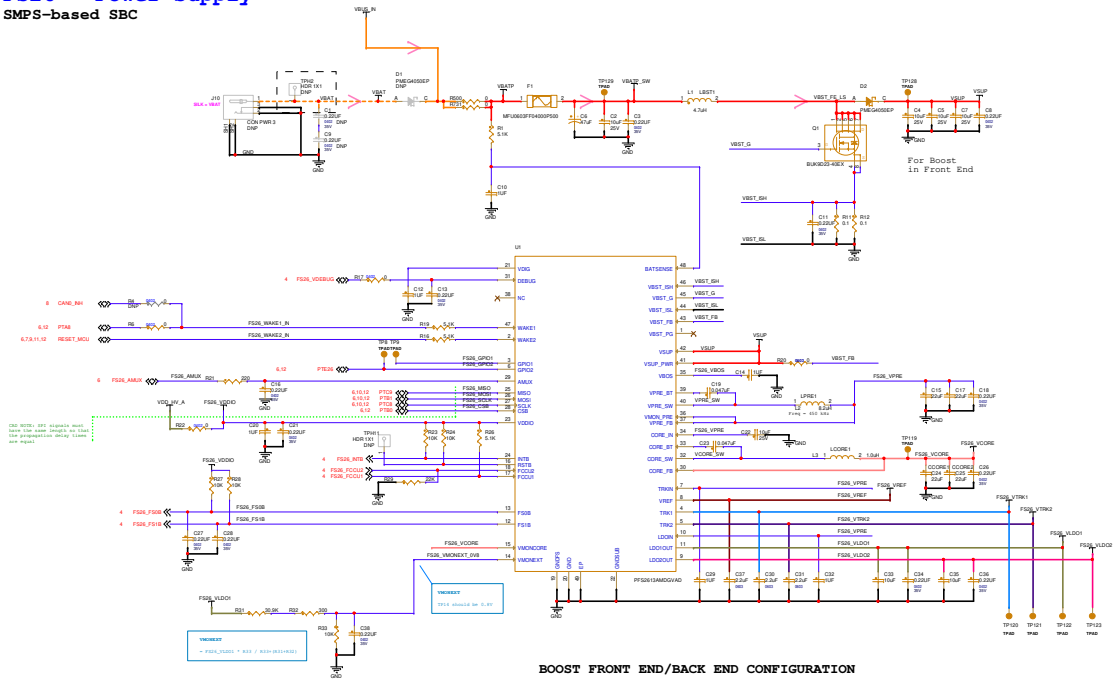
AUTO MCU GDL APPS	
DRAWING TITLE:	S32K3 Tlaloc – Block Diagram
TECHNICAL LEAD:	Efren Diaz
APPROVED:	Luis Garabito
DATE:	Thursday, July 25, 2024

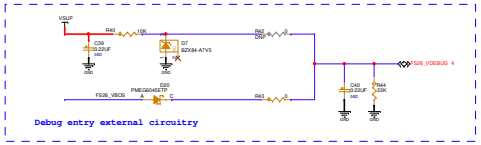
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Drawn by: Antonio Quiroz	Page Title: NOTES & BLOCK DIAGRAM	
Approved: Efren Diaz	Size C / Document Number SCH-94921 PDF: SPF-94921	Rev C
	Date: Wednesday, November 05, 2025 / Sheet 2 of 14	


VBUS DET
USB TYPE-C PD
Always ON LDO
USB PD
USB2.0 TYPE-C
CC OVP
SYS PWR SW
USB TYPE-C
Use Always On Power(with VBUS), avoid entering Dead Battery Mode.
I2C ADDR: 0x71(1110001x)
PTN5110
NX20P3483UK
NX20P0407
TLV76033DBZR
M096H-A020RT21A
PCMF1USB3S
Control and Gate Driver
Differential 90ohm
Difference tolerance 100mil
Pin C1 and C2 used for layout convenient
Control CC for Type-C PD power on/off control
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S32K344MINI-EVB
USB TYPE-C
SCH-94921 PDF: SPF-94921
Wednesday, November 05, 2025

FS26 – Power Supply

SMPS-based SBC

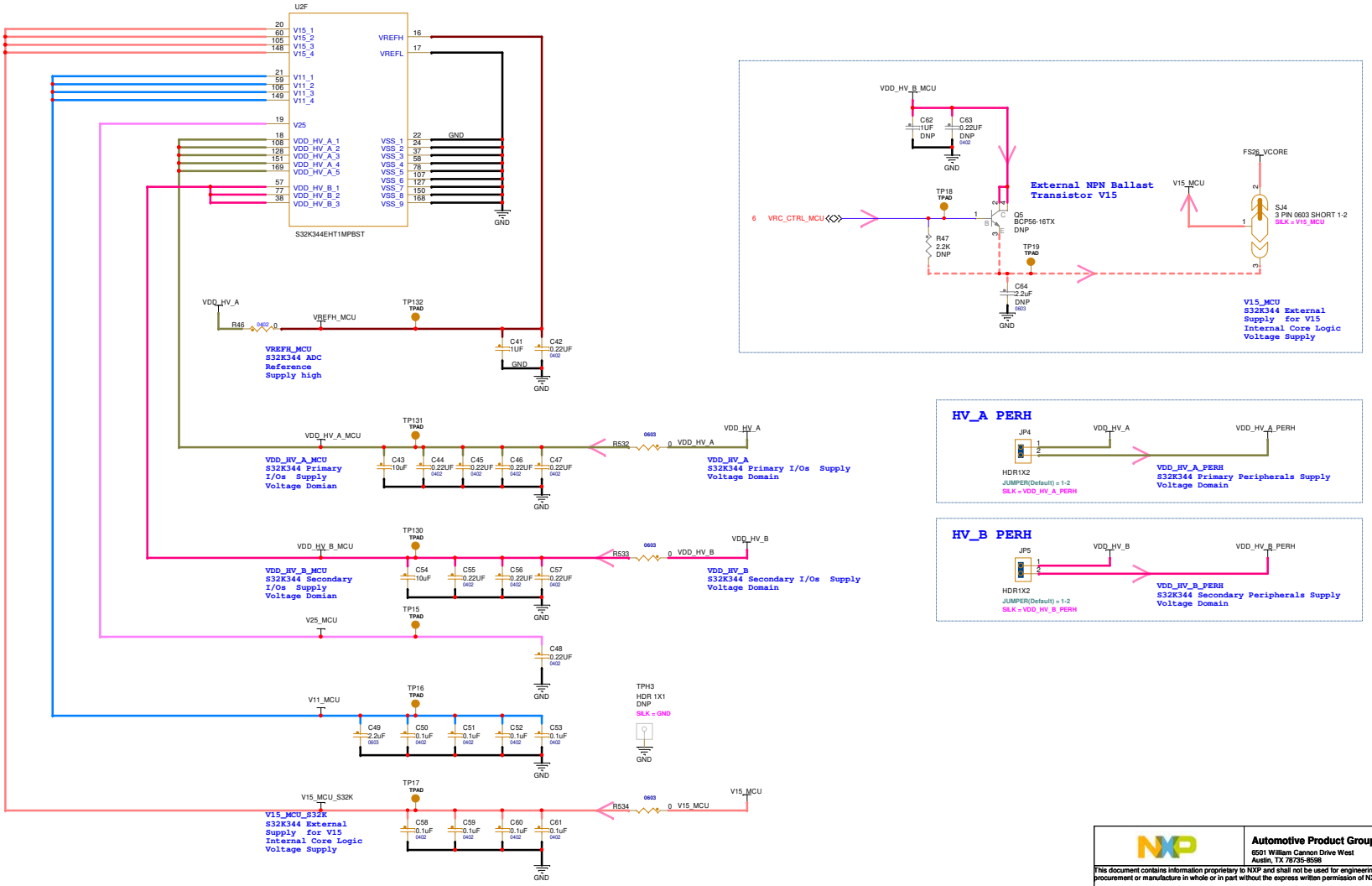





CAN2_INH
PTA8
RESET_MCU
FS26_WAKE1_IN
FS26_WAKE2_IN
FS26_AMUX
FS26_VDDIO
FS26_FS0B
FS26_FS1B
FS26_VCORE
FS26_VMONEXT_SVB
FS26_VLDO1
VMONEXT
VMONEXT should be 5.0V
VMONEXT = FS26_VLDO1 * R33 / R32+(R31+R32)

S32K3XX Microcontroller – 172HDQFP

Power Supply Pins



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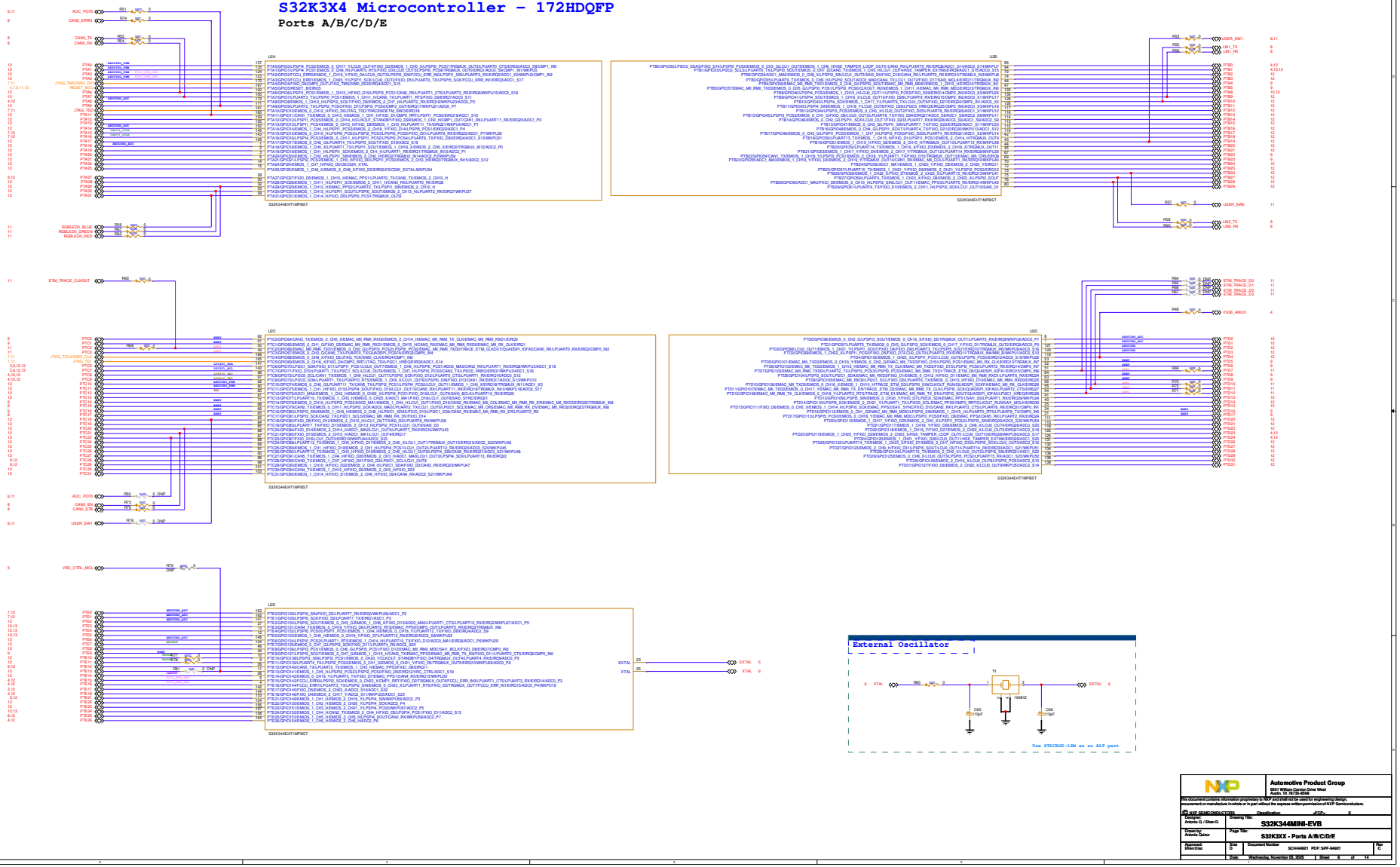
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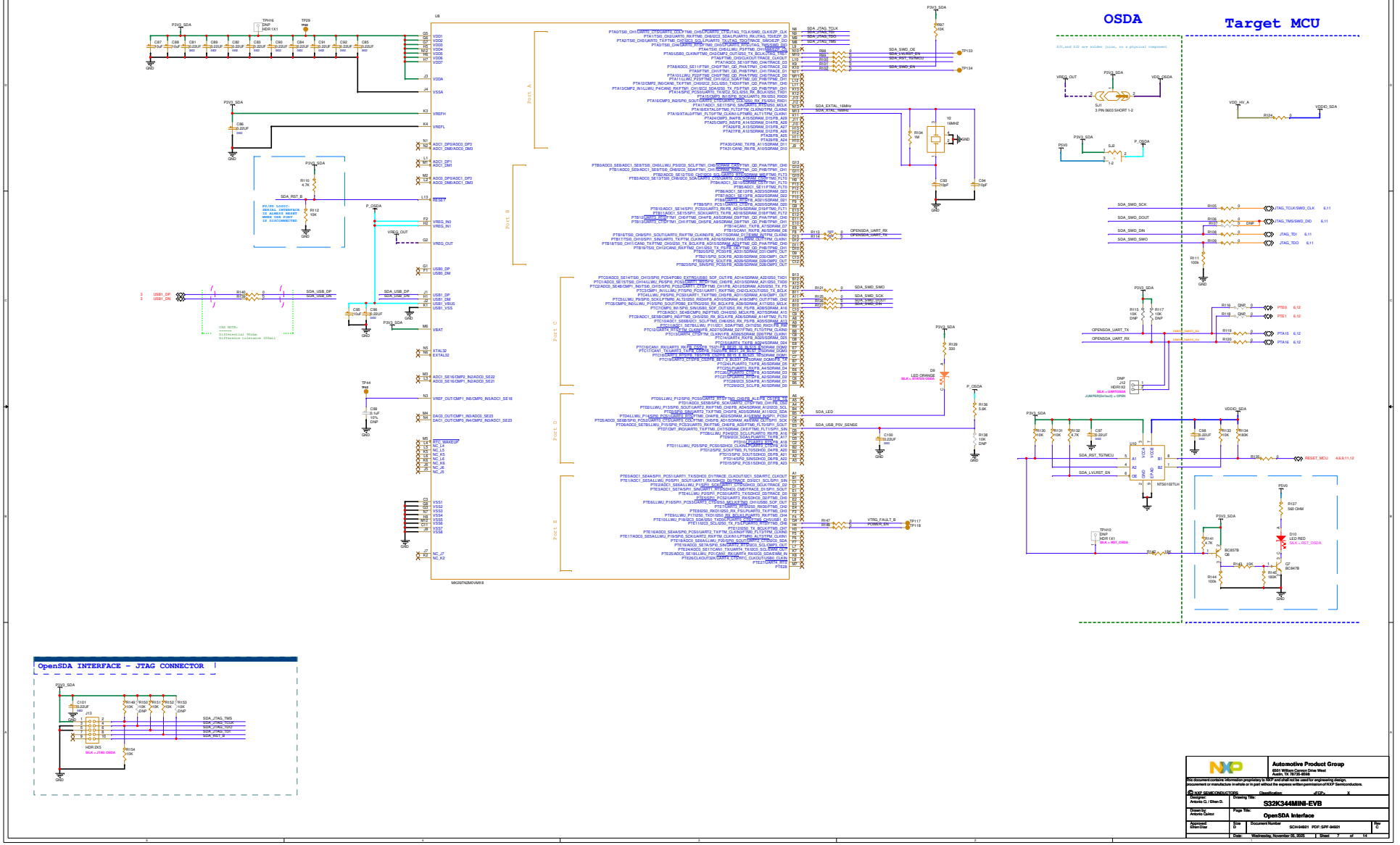
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Classification:	<FCP>	X	
Designer: Antonio Q. / Efren D.	Drawing Title: S32K344MINI-EVB		
Drawn by: Antonio Quiroz	Page Title: S32K3XX - MCU POWER Supply Pins		
Approved: Efren Diaz	Size: C	Document Number: SCH-94921 PDF: SPF-94921	Rev: C
Date: Wednesday, November 05, 2025	Sheet 5 of 14		

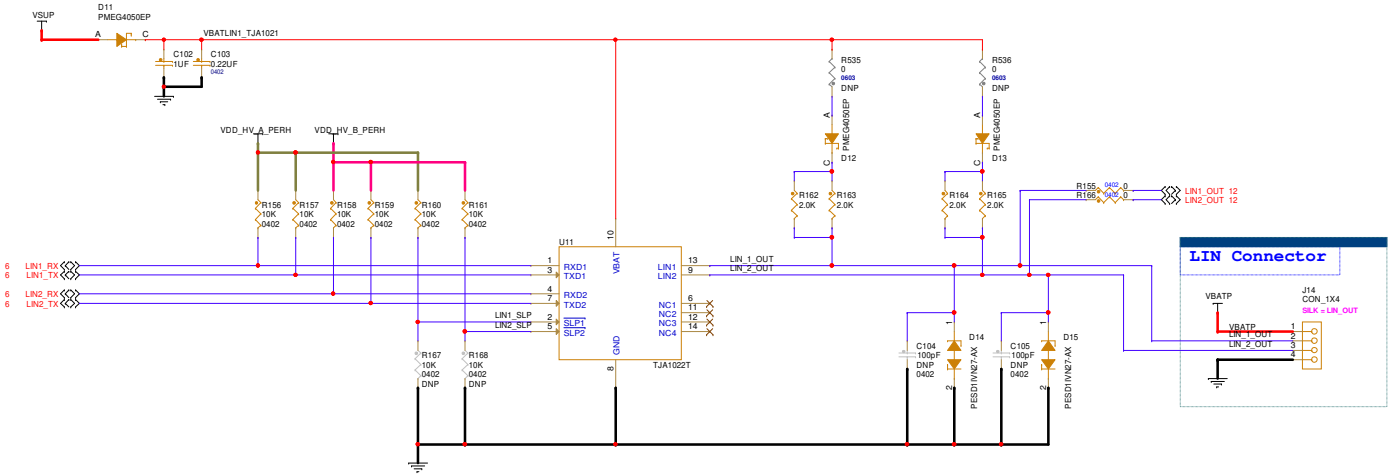
S32K3X4 Microcontroller – 172HDQFP

Ports A/B/C/D/E

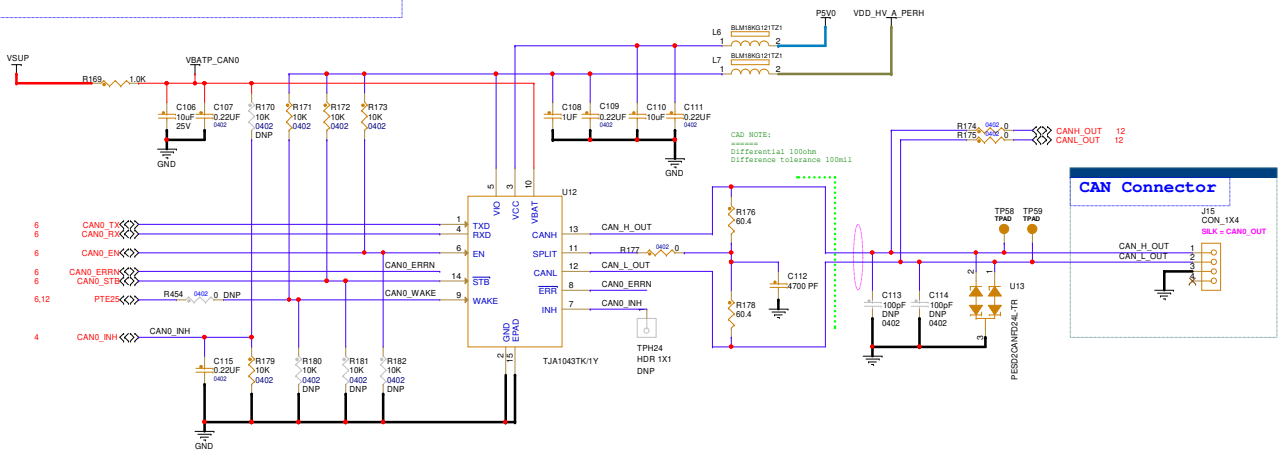



OSDA
Target MCU
OpenSDA INTERFACE - JTAG CONNECTOR
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S32K344MINI-EVB
OpenSDA Interface
SCH-94921
Wednesday, November 06, 2026

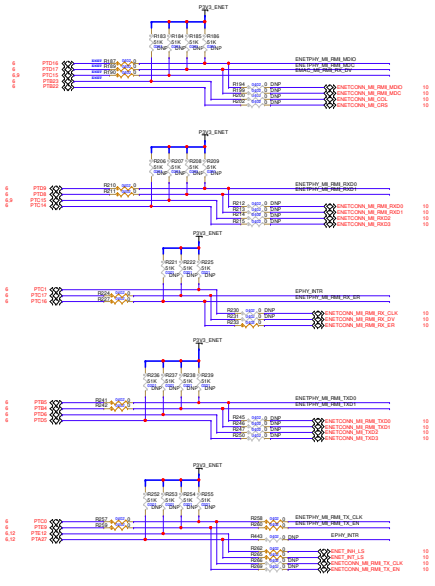
LIN Physical Layer
TJA1022T: Quad LIN 2.2A/SAE J2602 transceiver

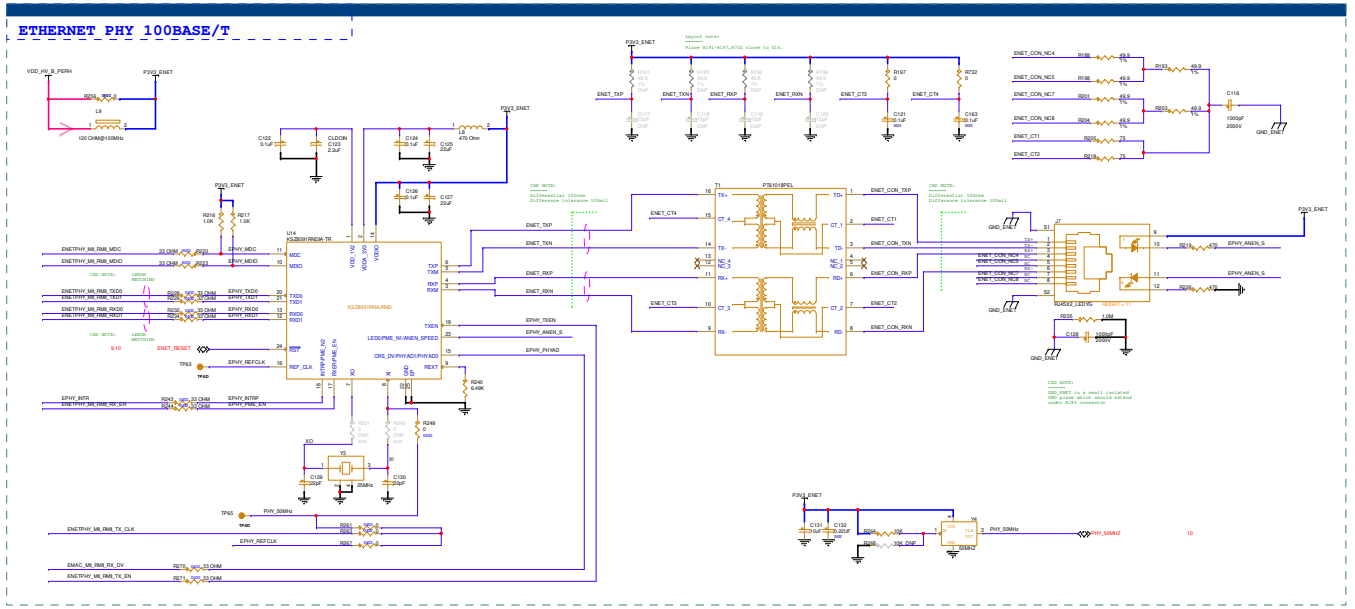


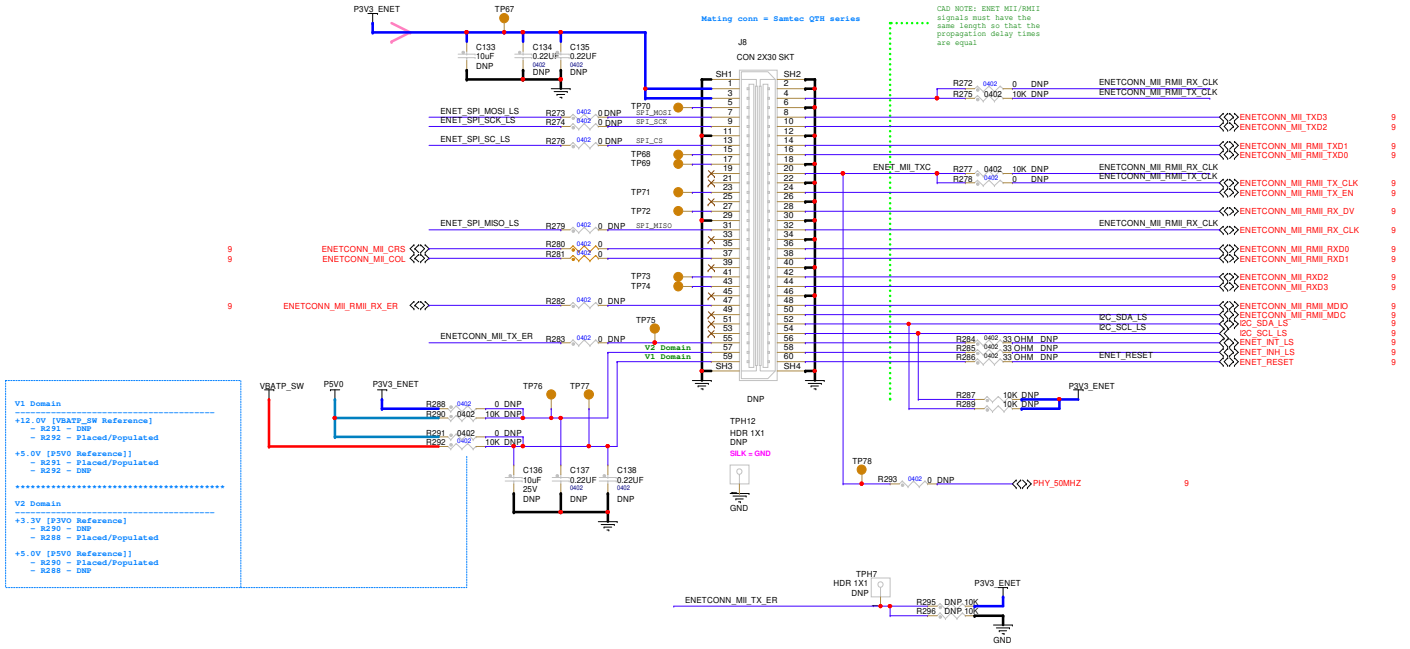
TJA1043 HS-CAN Transceiver



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Designer: Antonio Q. / Efren D.	Drawing Title: **S32K344MINI-EVB**	
Drawn by: Antonio Quiroz	Page Title: **CAN/LIN PHY**	
Approved: Efren Diaz	Size C / Document Number SCH-94921 PDF: SPF-94921	Rev C
	Date: Wednesday, November 05, 2025	Sheet 8 of 14




ETHERNET PHY 100BASE/T
KSZ8091RNDA-TR
P35/5 ENET
P35/6 ENET
P35/7 ENET
P35/8 ENET
P35/9 ENET
P35/10 ENET
P35/11 ENET
P35/12 ENET
P35/13 ENET
P35/14 ENET
P35/15 ENET
P35/16 ENET
P35/17 ENET
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P35/24 ENET



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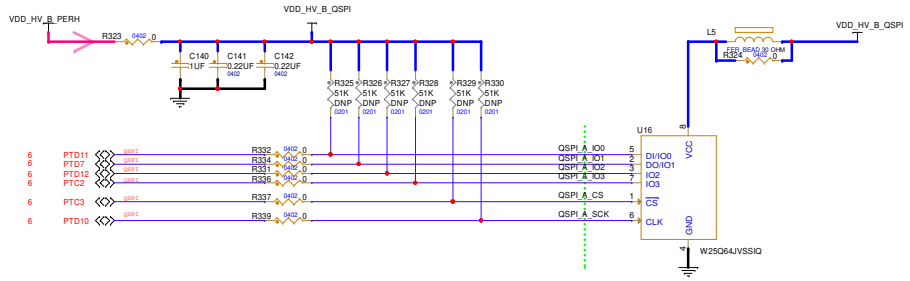
Drawing Title: **S32K344MINI-EVB**

Page Title: **ETHERNET CONNECTOR**

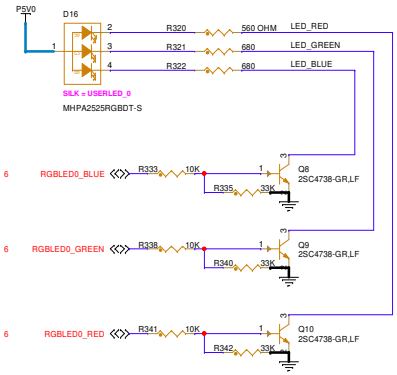
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C	SCH-94921 PDF: SPF-94921	C

Date: Wednesday, November 05, 2025 Sheet 10 of 14

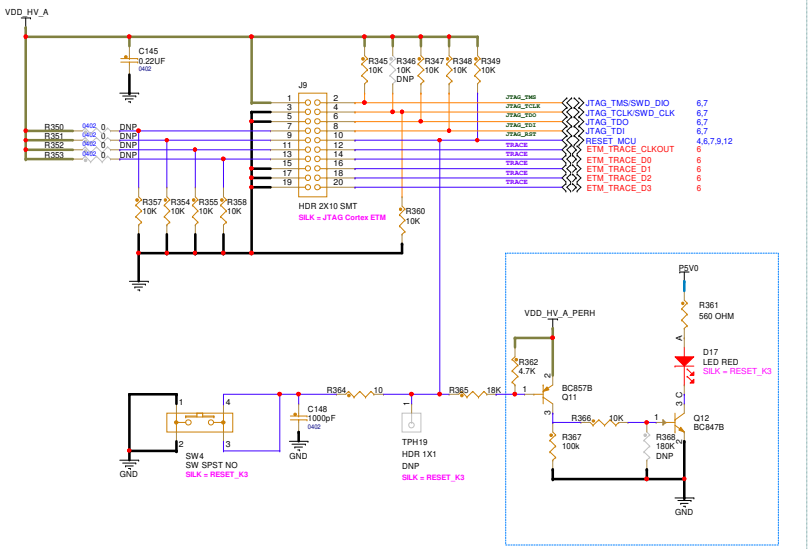
QSPI



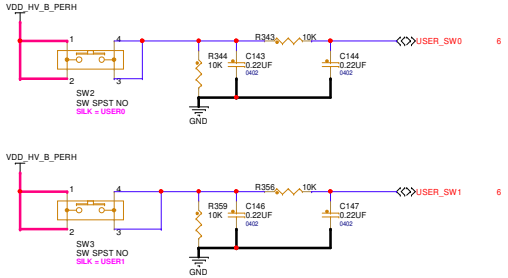
User RGB LED



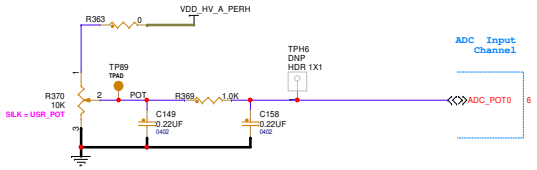
20-pin Cortex Debug D ETM connector



User Push Button - Active to HIGH/LOW



ADC Potentiometer



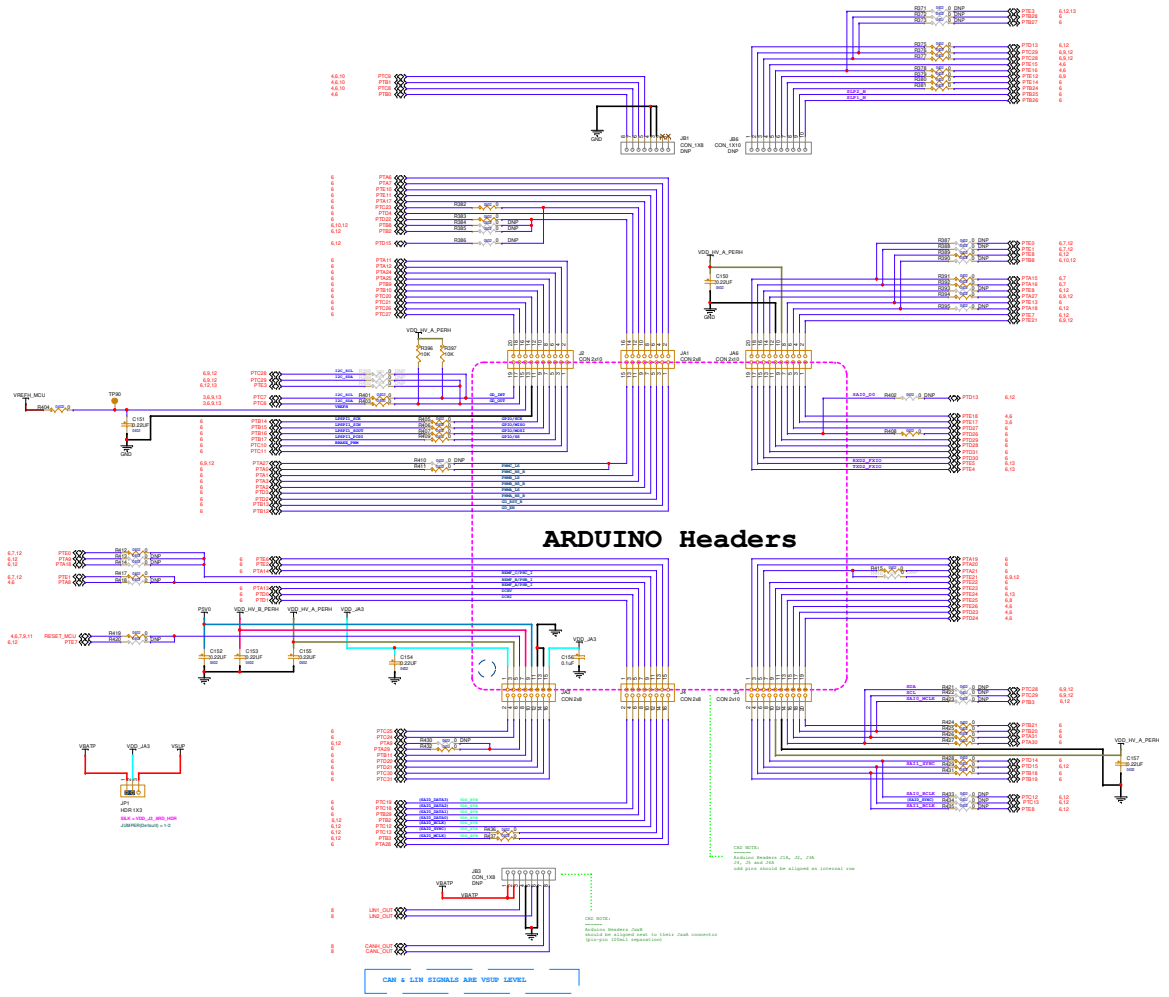
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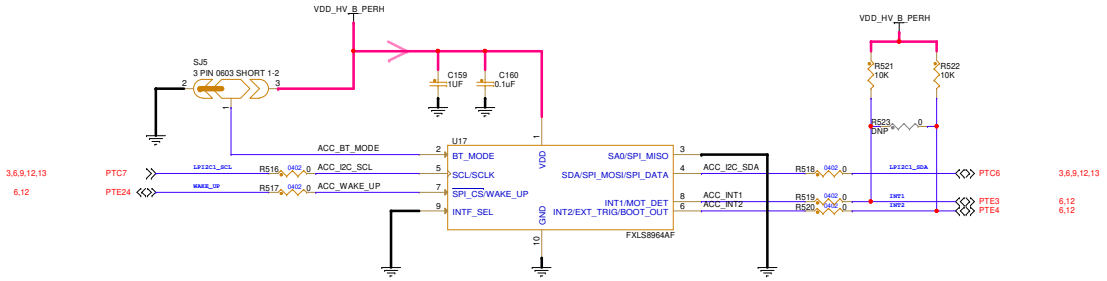
Designer: Antonio Q. / Efren D.	Drawing Title: S32K344MINI-EVB		
Drawn by: Antonio Quiroz	Page Title: QSPI Memory Interface		
Approved: Efren Diaz	Size C	Document Number SCH-94921 PDF: SPF-94921	Rev C
	Date: Wednesday, November 05, 2025	Sheet 11 of 14	



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Designer: Antonio G. / Efren D.	Checked by:
Drawn by: Antonio Cuevas	Page Title: ARDUINO Shield
Approved: Efren Diaz	Size: D; Document Number: SCH-94921 PDF:SPF-94921; Rev: C
	Date: Wednesday, November 08, 2006; Sheet 10 of 14

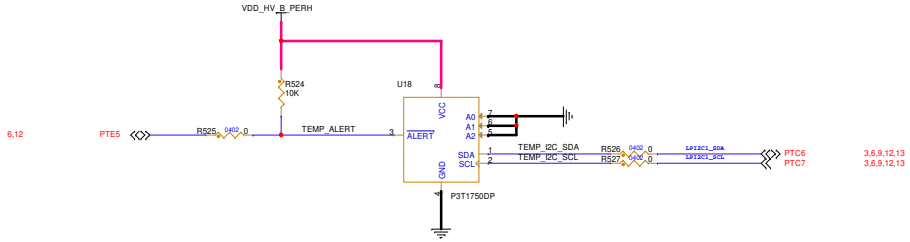
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ACCEL SENSOR



I2C ADDRESS: WR:30 RD:31

TEMP SENSOR



I2C ADDRESS: 0X90

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Drawn by: Antonio Quiroz	Page Title: SENSORS	
Approved: Efren Diaz	Size C / Document Number SCH-94921 PDF: SPF-94921	Rev C
	Date: Wednesday, November 05, 2025 / Sheet 13 of 14	

JUMPER Table

REF DES	JUMPER(DEFAULT)	PAGE NAME
JP11	1-2	04. FS26
JP4,JP5	1-2	05. MCU PWR
J12	OPEN	07. OPENSDA K26
JP1	1-2	12. ARDUINO HDRS

DNP Table

REF DES	ASSY_OPT	PAGE NAME
R88,R123,R128,R448,R449,R455, R456,R457,R458,R502,R730, TPH23	DNP	03. USB TYPE-C/PD
C1,C9,D1,J10,R4,R42,R529, TPH2,TPH11	DNP	04. FS26
C62,C63,C64,Q5,R47,TPH3	DNP	05. MCU PWR
R50,R64,R65,R66,R67,R76,R79, R81	DNP	06. MCU PORTS1
C99,J12,R107,R115,R116,R117, R118,R138,R150,R153,TPH10, TPH16	DNP	07. OPENSDA K26
C104,C105,C113,C114,R167, R168,R170,R180,R181,R182, R454,R535,R536,TPH24	DNP	08. CAN/LIN PHY
C117,C118,C119,C120,R183, R184,R185,R186,R191,R192, R194,R195,R196,R199,R200, R202,R206,R207,R208,R209, R212,R213,R214,R215,R221, R222,R225,R230,R231,R236, R237,R238,R239,R245,R246, R247,R248,R250,R251,R252, R253,R254,R255,R266,R268, R269,R305,R307,R308,R309, R311,R443,R537,R539,R542,R544	DNP	09. ETHERNET 10/100BASET
C133,C134,C135,C136,C137, C138,J8,R272,R273,R274,R275, R276,R277,R278,R279,R282, R283,R284,R285,R286,R287, R288,R289,R290,R291,R292, R293,R295,R296,R297,R298, R299,R300,R301,R302,R303, R304,TPH7,TPH12	DNP	10. ETHERNET CONNECTOR
R325,R326,R327,R328,R329, R330,R346,R350,R351,R352, R353,R368,TPH6,TPH19	DNP	11. QSPIA / JTAG / PERIPHERALS
JB1,JB3,JB6,R371,R372,R373, R384,R385,R386,R387,R388, R390,R393,R395,R398,R399, R400,R402,R410,R413,R414, R416,R418,R420,R421,R422, R423,R430,R433,R434,R435	DNP	12. ARDUINO HDRS
R523	DNP	13. SENSORS

SWITCH Table

REF DES	SWITCH(DEFAULT)	PAGE NAME
SW5	1-2 / 4-5	03. USB TYPE-C/PD

Shunt Table

REF DES	SHORT(DEFAULT)	PAGE NAME
SJ4	1-2	05. MCU PWR
SJ1,SJ2	1-2	07. OPENSDA K26
SJ5	1-2	13. SENSORS

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Drawn by: Antonio Quiroz	Page Title: NOTES & BLOCK DIAGRAM
Approved: Efren Diaz	Size C; Document Number SCH-94921 PDF: SPF-94921; Rev C
	Date: Wednesday, November 05, 2025; Sheet 14 of 14